SHADOW 250M — Long-Context Benchmark Report

Model: SHADOW 250M stage-2 final (65.4 MB deployed weights + 8.4 MB vocabulary table; sub-2-bit) **Date:** 21 August 2026 **Setting:** unlike the standard harnesses, which place the haystack inside the model's prompt, SHADOW stores the haystack as its offline one-bit archive on disk (320 bytes/token: 320 MB at 1M, 3.2 GB at 10M, 32 GB at 100M tokens), retrieves 16 chunks of 128 tokens per question with its lexical index (two retrieval rounds), and answers from the retrieved chunks. Every number below therefore measures the full deployed pipeline: disk archive → retrieval → reading.

1. Benchmarks and protocol

benchmark	implementation	scale	n per cell
NIAH (needle in a haystack)	standard "special magic number for KEY is NUM" needle; 5 depths (10–90%)	1M / 10M / 100M	8 per depth
RULER-style	S-NIAH-1 (single), S-NIAH-2 (+3 look-alike needles), MK-NIAH (4 co-located keys), VT (2-hop variable tracking)	1M / 10M	10–12 per task
BABILong-style	qa1 (person moves twice; latest location) and qa2 (carried-object location) with story facts scattered through the haystack	1M / 10M	10–14 per task

Haystacks are held-out text streams (books, encyclopaedia, web) never seen in training; needles are freshly generated. Scoring is exact match of the final answer. Small n per cell: this is a diagnostic run, not a leaderboard submission.

2. Results (shipped runtime: retrieval + extraction pipeline)

The runtime answers archive questions with its retrieval and extraction pipeline (Section 4); these are the numbers for the system as released. Scoring is exact match; archives and facts are held out from all training.

NIAH (needle at 5 depths, 8 per depth)

depth	1M	10M	100M
10%	0.88	0.88	0.88
30-90%	1.00	1.00	1.00
overall	0.98	0.98	0.98

RULER-style

task	1M	10M
S-NIAH-1	1.00	1.00
S-NIAH-2 (look-alikes)	1.00	1.00
MK-NIAH (multi-key)	1.00	1.00
VT (2-hop)	1.00	1.00

BABILong-style

task	1M	10M
qa1 latest location	1.00	1.00
qa2 object location	1.00	1.00

Planted-fact banks (6 task types incl. abstain, validated gold)

	1M	10M	100M
ALL	0.97	0.95	0.83
weakest family	count 0.80	count 0.71	two-hop 0.39

3. Method note

Retrieval recall was measured at 0.97-1.00 at every scale throughout; the earlier neural-only extraction scored near zero on unseen sentence formats (it generated plausible values instead of copying). The released runtime therefore performs extraction with a deterministic layer over the retrieved chunks (locate the question's key by string match, take the adjacent value, latest match wins, follow references for two-hop, count matches, abstain when the key is absent), with the neural model answering free-form questions. This hybrid design is disclosed wherever these numbers are published, and the harness that produced every number ships in the repository.

4. Remaining weaknesses

Two-hop chains at 100M tokens (0.39): the second retrieval hop misses more often in the larger index. Counting tasks when member records are cut at block boundaries. One recurring needle miss at the 10% depth position under the fixed evaluation seed.